

Dnmt1-Independent CG Methylation Contributes to Nucleosome Positioning in Diverse Eukaryotes

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Evidence abstract

Dnmt1 epigenetically propagates symmetrical CG methylation in many eukaryotes.

Their genomes are typically depleted of CG dinucleotides because of imperfect repair of deaminated methylcytosines.

Here, we extensively survey diverse species lacking Dnmt1 and show that, surprisingly, symmetrical CG methylation is nonetheless frequently present and catalyzed by a different DNA methyltransferase family, Dnmt5.

Numerous Dnmt5-containing organisms that diverged more than a billion years ago exhibit clustered methylation, specifically in nucleosome linkers.

Clustered methylation occurs at unprecedented densities and directly disfavors nucleosomes, contributing to nucleosome positioning between clusters.

Dense methylation is enabled by a regime of genomic sequence evolution that enriches CG dinucleotides and drives the highest CG frequencies known.

Species with linker methylation have small, transcriptionally active nuclei that approach the physical limits of chromatin compaction.

These features constitute a previously unappreciated genome architecture, in which dense methylation influences nucleosome positions, likely facilitating nuclear processes under extreme spatial constraints.

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